

Isidora Vidal Prieto

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Geographer with hands-on experience in satellite imagery analysis and cloud-native geospatial pipelines, working with Sentinel-2 and Landsat data through Google Earth Engine (JavaScript and Python APIs), Python (Dask, xarray, odc-stac), and R. Experienced in applied spatial statistics (Cox proportional hazards models, time series analysis) and building geospatial data pipelines for environmental and urban research. While my background is stronger in remote sensing and statistical modeling than in deep learning frameworks (TensorFlow, PyTorch, Scikit-Learn) or spatial databases (PostGIS), I bring a solid foundation in Python-based geospatial analysis and a strong interest in growing into spatial machine learning applications.

PROFESSIONAL EXPERIENCE

Technical Collaborator — Data Cube Chile — Universidad Adolfo Ibáñez (UAI)

Santiago, Chile · May 2026 – Present

- Adapted Digital Earth Australia (DEA) notebooks to the EASI/JupyterHub environment of Data Cube Chile, covering change detection (ENDISI/NDVI), burn area mapping, seasonal water extent monitoring (MNDWI), and fuel moisture content (FMC).
- Designed cloud-native geospatial pipelines in Python, Dask, and xarray over Sentinel-2 and Landsat 8/9 data, including cloud masking, spectral index computation, and output as Cloud-Optimized GeoTIFF (COG).
- Validated and visualized results with Google Earth Engine (JavaScript and Python APIs) and maintained technical documentation with version control via Git/GitHub, contributing to an institutional repository (UAI-CTE/deamericas-notebooks).

Technical Collaborator — FONDECYT Iniciación Grant N°11240322 — Pontificia Universidad Católica de Chile

Santiago, Chile · April 2026 – Present

- Applied Cox proportional hazards models to assess how urban environment characteristics modify the association between extreme heat and preterm birth across 33 communes of the Santiago Metropolitan Region (2009–2020).
- Integrated and systematized environmental and sociodemographic datasets (national population data, social vulnerability, vegetation cover, road and industrial density) for spatial analysis in R.
- Standardized and cleaned data following project methodological protocols (commune name normalization, variable coding, handling of missing and non-comparable data across years).

GIS Analyst — Centro de la Tierra y el Espacio — Coastal GIS Project

Chile · March 2026 – Present

- Produced spatial inventories and regional technical fact sheets for Chile's coastal zones, supporting national geospatial integration.
- Provided fieldwork and GIS analysis support as part of an interdisciplinary team, including processing, systematization, and quality control of coastal spatial data.
- Developed thematic cartography and technical reports to communicate findings to diverse audiences.

Technical Collaborator — Centro de la Tierra y el Espacio — Magallanes Peatland Project

Tierra del Fuego, Chile · May 2026 – Present

- Detected degraded/cut peatlands and beaver intrusion in Tierra del Fuego through land-cover classification (CMM) and spatial analysis in R (terra package).
- Generated and validated ~186 sampling points using GIS and Google Earth Pro visual verification, with systematic coding of cut and degraded sites.
- Maintained data traceability through field tracking spreadsheets (Excel) and layers exported in KML format.

ADDITIONAL EXPERIENCE

Geospatial Data Quality Analyst — Raster4 *(Santiago, Chile · March 2026)*

Validated consistency between satellite imagery and digital surface models (DSM); identified, classified, and logged incidents for geospatial data quality control.

Research Assistant — FONDECYT Iniciación Grant N°11220701 *(Santiago, Chile · April 2024 – October 2025)*

Processed satellite imagery (Landsat, Sentinel-2) to detect glacier extent change (1985–2023) through remote sensing and GIS; analyzed climate and glaciological data using regressions, correlations, and time series.

Laboratory Assistant — Dendrochronology Laboratory, UMAG *(Punta Arenas, Chile · March – July 2024)*

Built tree-growth chronologies (*Nothofagus pumilio*) for climate variability analysis; participated in field campaigns for sensor installation and ecophysiological monitoring.

SKILLS

Cloud geospatial: Open Data Cube, Dask, xarray, odc-stac, Google Earth Engine (JS/Python), Sentinel-2, Landsat, COG format, cloud raster processing

Remote sensing: Supervised classification, change detection, spectral indices (NDVI, NDWI, NDII, ENDISI), cloud masking

GIS & cartography: QGIS, ArcGIS Pro, Google Earth Pro, spatial analysis, thematic cartography

Programming & data: Python, R, JavaScript, Git/GitHub, Quarto

Statistics: Cox models, regressions, time series, correlations, spatial statistics, predictive modeling

Other tools: Excel/SPSS, database management and systematization, technical report writing

EDUCATION

B.A. in Geography — Universidad de Chile

Santiago, Chile · March 2020 – August 2025

- Distinction: Varsity Athlete.
- Thesis: "Reconstructing Climate Variability and Its Relationship to Glacier Change – Lanín Volcano" (Universidad de Chile Repository), focused on dendrochronology, remote sensing, and statistical methods.

Bioengineering (incomplete) — Universidad de Concepción

Concepción, Chile · March 2017 – August 2019

LANGUAGES, CERTIFICATES & ACTIVITIES

- Languages: Spanish (native, C2) · English (fluent, C1)
- Certificate: Introduction to Spatial Data Infrastructures (SDI)
- Other: ISFiT (International Student Festival in Trondheim), Norway (Feb. 2023), international discussion panels on globalization and climate change.
- Club Andino Universitario, Santiago (Sept.–Nov. 2025) , training in wilderness orientation, mountaineering, and first aid.
- Class B driver's license.